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Infinitesimal Analysis in the Engineer's Education

Brahadeesh Sankarnarayanan

Every engineering student has, at some point in their coursework, done something that their analysis professor would call slightly illegal. When they encounter a separable differential equation like $dy/dx=xy$, they cheerfully separate the variables, $dy/y=x\,dx$, integrate both sides, and arrive at the correct answer. When they invoke the chain rule to differentiate a composite function $h(x)=g(f(x))$, they do so as though $dh/dx=dg/df\ df/dx$ simply by cancelling the "df" terms. In a math course, the instructor would emphasize that these are just mnemonics: the *true* justification involves formalizing limits via ε-δ arguments, and that the differentials dy and dx are convenient notation without any independent mathematical existence.

Yet, if the method always yields the correct answer, perhaps what we have is more than a mnemonic?

The answer is positive, as was shown by Abraham Robinson in the 1960s in a stunning piece of work, laying down the foundations for what is called *nonstandard analysis*.

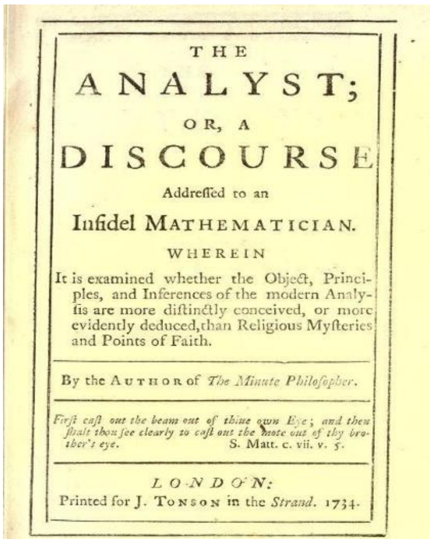

Figure 1. Abraham Robinson (c. 1970)

A brief history of infinitesimals

When Isaac Newton and Gottfried Wilhelm Leibniz independently formulated the calculus in the seventeenth century, both worked freely with infinitesimals: quantities smaller than every positive real number in magnitude, and yet not equal to zero. Of course, neither man could give a completely satisfactory description of what an infinitesimal really is. To a large extent, the effectiveness of the language in describing reality was its own proof. The calculus worked, and produced correct answers. In the hands of masterful analysts such as Leonhard Euler, the calculus led to explosive developments over the next century.

Yet, the problems were not entirely brushed under the carpet. Perhaps the most famous criticism came from the Anglo-Irish philosopher and Bishop of Cloyne, George Berkeley, who published in 1734 *The Analyst* [1], a scathing critique of the foundations of calculus in which he memorably described infinitesimals as "the ghosts of departed quantities": the mathematician invokes a quantity such as dx in a calculation, ostensibly a nonzero quantity, but then discards it at the end as if it were zero! What justifies this?


Figure 2. Bishop George Berkeley


THE
ANALYST;
OR, A
DISCOURSE
Addressed to an
Infidel MATHEMATICIAN.
WHEREIN
It is examined whether the Object, Principles, and Inferences of the modern Analysis are more distinctly conceived, or more evidently deduced, than Religious Mysteries and Points of Faith.

By the AUTHOR of *The Minute Philosopher*.

First cast out the beam out of thine own Eye; and then shalt thou see clearly to cast out the mote out of thy brother's eye. S. Matt. c. vii. v. 5.

LONDON:
Printed for J. TONSON in the *Strand*. 1734.

Figure 3. Frontispiece of The Analyst

It took nearly another hundred and fifty years for mathematics to answer Berkeley's objection. The work of Cauchy and Weierstrass in the nineteenth century formalized the ε-δ analysis, and this became the standard rigorous foundation for calculus ever since.

Robinson's NSA

Then, in 1966, Abraham Robinson published a book titled *Non-standard Analysis* [2].

Using techniques from mathematical logic (specifically, model theory), Robinson rigorously constructed an extension of the real numbers in which infinitesimals exist as genuine mathematical objects, with all their intuitive properties preserved and with the manipulations of the seventeenth-century calculus given a natural universe to live in.

This extended number system is called the *hyperreal numbers*, denoted *R. It contains all the ordinary real numbers, but also two new kinds of quantities. First, there are *infinitesimals*: positive numbers smaller than every positive real, and their negatives. If ε is a positive infinitesimal, then $\varepsilon<1/n$ for every natural number n. Second, there are *unlimited hyperreals*: the reciprocals of infinitesimals, larger than every real number.

Every finite hyperreal x has a unique *standard part*, denoted st(x): the unique ordinary real number that x is infinitesimally close to. So $st(3+\varepsilon)=3$ whenever ε is infinitesimal. The standard part is the bridge between the extended system and the ordinary reals. The standard part operator at last justifies the action of dropping the infinitesimal part at the end of calculations, which Berkeley criticized all those centuries ago.

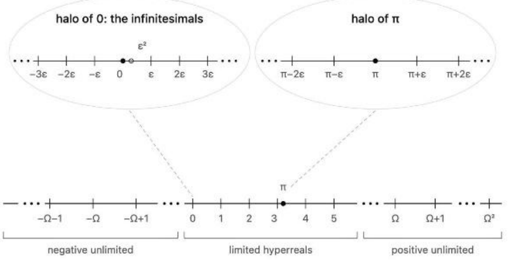


Figure 4. Schematic of the hyperreal line showing the limited and unlimited hyperreals, the infinitesimals, and the halo of hyperreals infinitely close to π.

One of the critical properties of *R which justifies the entire theory is the *transfer principle*. Roughly speaking, this principle guarantees that any first-order mathematical statement that is true of the ordinary real numbers remains true of the hyperreals. The hyperreals form an ordered field. Every nonzero hyperreal has a reciprocal. The transferred sine function still satisfies $\sin^2(x)+\cos^2(x)=1$. And so on. The transfer principle is what ensures that calculations with infinitesimals never lead one astray.

Three examples

Let me now demonstrate what nonstandard analysis looks like in practice, through three examples from the undergraduate classroom that reveal its intuitive character and pedagogical potential.

The derivative of x^2**.** In the standard ε-δ treatment, computing this derivative requires several careful steps: one considers the difference quotient $(f(x+h)-f(x))/h$, argues that its limit as $h \to 0$ exists, and demonstrates that this limit equals $2x$. The argument is careful but somewhat opaque; students must simultaneously perform algebra on the quotient and reason about the limiting process.

In the hyperreal calculus, the entire computation reduces to a single algebraic manipulation. Let ε be a nonzero infinitesimal. Then $((x+\varepsilon)^2-x^2)/\varepsilon = (2x\varepsilon+\varepsilon^2)/\varepsilon = 2x+\varepsilon$. The derivative of x^2 at the point x is now, *by definition*, the standard part of this quantity: $\mathrm{st}(2x+\varepsilon)=2x$.

Notice what this computation has done. It has separated two conceptually distinct steps that the ε-δ argument entangles: the algebraic step of simplifying the difference quotient, and the analytical step of resolving what happens as the increment shrinks. In the nonstandard approach, one first performs pure algebra on a hyperreal expression, and only at the end takes the standard part. The pedagogical benefit is that students see clearly what calculus is really asking — namely, what does the difference quotient become when the increment is infinitesimally small? — without the question being obscured by the machinery of limits.

Continuity. A second illustration comes from the definition of continuity. The standard textbook definition reads as follows: *a function* f *is continuous at a point* a *if, for every* $\varepsilon>0$, *there exists a* $\delta>0$ *such that* $|x-a|<\delta$ *implies* $|f(x)-f(a)|<\varepsilon$. This definition has tripped up generations of students, for whom the sequence of alternating quantifiers is (quite understandably!) a tangled web of varying quantities, instead of giving the concept any explanatory power.

The nonstandard definition reads: *a function* f *is continuous at a point* a *if* $\mathrm{st}(f(a+\varepsilon))=f(a)$ *for every infinitesimal* ε. In ordinary language: if you nudge the input by an infinitesimal amount, the output is nudged by at most an infinitesimal amount. This is precisely the intuition that every engineer and physicist has always had about continuity. The intuitive definition is now the formal one.

The pedagogical advantage is even sharper for the notion of *uniform* continuity. In the standard treatment, uniform continuity differs from ordinary continuity by the order in which the quantifiers appear — for every ε, there exists a *single* δ that works uniformly for all points in the domain — which is a notoriously subtle distinction to get across. In the nonstandard treatment, f is uniformly continuous on an interval I if the same relation $\mathrm{st}(f(x+\varepsilon))=f(x)$ holds for every *hyperreal* x in the hyperreal extension *I of the interval. The quantifier shift has been replaced by a geometric shift: uniform continuity of a real function is tested by the same continuity criterion, but at a finer set of points.

The chain rule. The third example concerns the chain rule, and it is perhaps the most vivid vindication of the engineer's intuition. If z is a differentiable function of y, and y is a differentiable function of x, then $dz/dx=dz/dy\ dy/dx$.

The genius of Leibniz's notation is that this identity feels like a natural consequence of "canceling" dy on the right-hand side. But this cancellation is emphatically not allowed in the standard framework, because dy/dx is not a ratio at all. It is just a notational shorthand for the derivative. The proof instead requires a delicate argument that constructs an estimate matching what the cancellation would have produced. Many introductory courses omit the proof entirely on the grounds that it is fiddly.

In the nonstandard framework, the cancellation is real. The quantities dz, dy, dx are actual infinitesimals — the increments in each variable that arise when x is nudged by an infinitesimal dx. The chain rule becomes nothing more than the arithmetic of these infinitesimal fractions, valid up to standard part by direct algebra. The mnemonic that the student was told is "just notation" is, in Robinson's framework, the proof itself.

Infinitesimals in the curriculum?

A natural question presents itself. If nonstandard analysis provides cleaner definitions, simpler proofs, and better alignment with intuition, then why has it not become the standard way of teaching calculus?

In fact, the advantage of NSA in the teaching of calculus was apparent to a number of mathematicians and educators following the publication of Robinson's work.
As early as 1976, the logician H. Jerome Keisler published *Elementary Calculus: An Infinitesimal Approach*, a full calculus textbook built entirely on nonstandard analysis. It is rigorous, careful, and remarkably readable. Since 2002, Keisler has made it freely available online [3]. It has been used in a number of courses around the world, and by all accounts it has been pedagogically successful. However, the change has not caught on in most curricula.

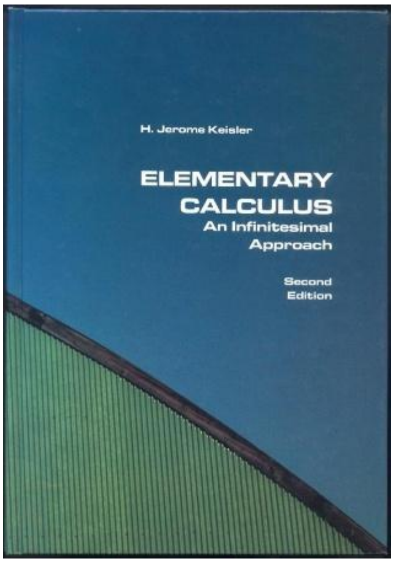


Figure 5. Cover of Keisler's Elementary Calculus textbook

There are several reasons that explain the cautious reception of NSA.

Firstly, the construction of the reals from the rationals by Dedekind cuts or Cauchy sequences is a standard procedure that every mathematician is familiar with, and which a diligent undergraduate can follow with minimal guidance. The hyperreals, however, are constructed via ultrafilters, which demands considerably more logical machinery. An instructor who is unprepared to answer questions on the construction themselves would not be in a position to carefully guide students to use the hyperreals.

Secondly, by the time Robinson's work appeared in the 1960s, the ε-δ formalism had been the meaning of "rigorous calculus" for nearly a century. Textbooks, examinations, syllabi, and the training of professors themselves were all built on this definition of the limit. Changing the curriculum is easier said than done, particularly if the framework produces benefits perceived to be marginal rather than transformative. For instance, NSA by itself does not prove theorems which the standard machinery cannot do so on its own, at least in principle.

Thirdly, it is a curious scenario that engineers and physicists who routinely work with the language of infinitesimals rarely have a need for a rigorous framework as well, whether it be the ε-δ one or NSA. Similarly, mathematicians who have little interest in model-theoretic foundations are largely uninterested in learning a new language to express their analytic intuitions, since the ε-δ framework suffices to write rigorous proofs regardless of the method by which the results were arrived at.

What next?
None of this is to argue that we should rewrite the calculus curriculum. My purpose is more modest: I want to suggest that the engineering student who manipulates dy and dx as though they were ordinary quantities was never doing anything mathematically wrong, and it is a disservice to both the engineer and the mathematician to suggest that these intuitions cannot be given proper meaning. A one-size-fits-all approach to a geometrically rich subject such as calculus, or mathematical analysis in general, is unnecessary in a world enriched by Robinson's NSA. Perhaps when a student finds that their infinitesimal reasoning leads them consistently to correct answers by "wrong" methods, they can be told that there is in fact a hyperreal world nearby, waiting to be explored — and perhaps the instructor can open Keisler's book alongside them!

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